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Wellbore Integrity Impact on Carbon Leakage to Ensure Safe Geological Sequestration

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Abstract

Secure and safe geological sequestration is critical for the successful deployment of carbon capture and storage (CCS) projects and to mitigate global warming. Oil/gas reservoirs and saline aquifers, that are suitable for CO_2 sequestration, often have older pre-existing wells near the potential CO_2 injection wells. The legacy producing and abandoned wells have the potential to behave as CO_2 leakage pathways, despite the existence of a good seal formation or structural certainty. In this study, this poorly understood and largely unknown leakage potential is researched to quantify the carbon leakage rate along one legacy well when carbon is injected into a storage formation in the Denver-Julesburg (DJ) Basin. This study investigates the risk of containment loss for a leaky well using a one-square-mile section of the DJ Basin. The results show that the completion of an older wellbore has a significant effect on the leakage rate results. A small portion (10^{-8}) of the injected carbon can leak via a multi-segmented wellbore, while around 2.5% CO_2 can leak to the drinking water layer via an open wellbore. Our results indicate that for secure carbon sequestration, a good geological storage site should be kept away from wells that have a surface casing depth of 476 ft or shallower and were drilled before 1994 in Colorado. The effective well permeability highly impacts the leakage rate, implying the importance of estimating the most representative value/range for this parameter. This risk assessment research is a part of the pre-feasibility study to support potential large-scale geological carbon storage operations in Colorado.

Introduction

Safe and secure carbon storage is an integral component to mitigate climate change. It is a process of injecting CO_2 into the subsurface and thereby reducing the anthropogenic CO_2 emissions. Multiple pilot-scale and large-scale subsurface carbon storage projects in the U.S. and globally have demonstrated this technology's feasibility. For instance, the In Salah carbon capture and storage project in central Algeria successfully stored over 3.8 million tons of CO_2 in the Krechba formation during year 2004 to 2011 (Ringrose et al., 2013).

Along with the wide deployment of carbon storage, there is an urgent need for assessing storage performance. After CO_2 is injected into a geological formation, when the CO_2 plume spreads and encounters the existing wellbores, CO_2 may leak through various pathways surrounding the well casing (Gasda et al.,

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2007). Due to defects caused by either poor well completion or subsequent damage arising from chemical reactions and thermal and/or mechanical stresses, wells are often considered as potential leakage conduits in a CO_2 sequestration site (Gasda et al. 2004a, 2004b). The possible leakage paths in an abandoned well include flow pathways along rock-cement and casing-cement interfaces, as well as through the degraded materials or materials improperly formed during the plugging process. During CO_2 injection, the reservoir stress change and deformation will impact the hydraulic aperture of rock-cement and casing-cement interfaces, which are the most likely pathways for potential CO_2 leakage around existing wells (Bachu and Bennion 2009; Tau et al. 2011). Additionally, the fractures formed in the cement may be influenced by chemical degradation once exposed to CO_2 (Huerta et al. 2009; Wigand et al. 2009). In the In Salah CO_2 injection project, a small CO_2 leakage is observed along an old well that was drilled in the 1980s and not cemented across the interval of the CO_2 injection zone (Mathieson et al., 2011; Rutqvist, 2012). This highlights the risk of pre-existing wells.

To estimate risks associated with geological carbon storage, the NRAP (National Risk Assessment Partnership) tools have been collaboratively developed by five DOE (Department of Energy) national laboratories (NETL, 2021). The NRAP-Open-IAM (Integrated Assessment Model) tool used in this study presents a modeling approach that can efficiently couple reservoir and wellbore components with low computation cost (Pawar et al. 2016). Two wellbore model components, the open wellbore model (Bacon et al. 2021) and the multi-segmented wellbore model, are studied to estimate the leakage rate during and after the carbon injection operations. Moreover, the effective permeability of the "disturbed zone" outside the well casing is an important parameter in leakage models. However, there has been little research about the reasonable range of this parameter. This topic will also be researched in this paper.

Hydrocarbons in DJ Basin have been producing since the 1970s. Further, the more recent multistage hydraulic fracturing stimulation has made the low-permeability and low-porosity formations such as Niobrara through Greenhorn active target zones. In 2020, around 25,734 wells produced little to no oil in Colorado (Jaffe, 2021). Such wells can create potential leakage risk pathways for future carbon storage. Here we study this problem using over a one-square-mile section in the Wattenberg Field of the Denver-Julesburg (DJ) Basin located in Colorado. The goal is to quantify the leakage rate when CO_2 escapes through the well that is 1100' away from the CO_2 injector. To start with, a reservoir simulation model was built to reflect the pressure and saturation changes during the 30-year injection period and the 100-year post-injection period. This model, based on an ongoing field study, includes the CO_2 injector, the legacy well which is the potential pathway of CO_2 leakage, and the carbon storage formation. Together with the properties of USDW (underground sources of drinking water), shale zones, aquifers, and seal formations, the simulation results from the reservoir were input into the NRAP-Open-IAM tool to quantitatively assess the carbon leakage rate into the overburden.

Study Area

CCUS potential in Colorado

The top-ranked CO_2 emission facilities in Colorado are located above the DJ Basin. According to the carbon emission data (EPA Flight, 2022), 13 facilities in Colorado emit more than 0.7 million tons of CO_2 in the year 2020, among which there are 5 coal power plants. All these coal plants are scheduled to be retired by 2034 (Colorado Energy Office, 2022). When excluding these coal plants, the locations of the top 8 CO_2 producers are annotated in the map of Colorado, as shown in Fig. 1. All the 8 CO_2 emission facilitates are located in the DJ Basin, whereas half are located in the Wattenberg Field (marked by the green dashed curve). This makes DJ Basin, especially the Wattenberg field, a good target for carbon storage.

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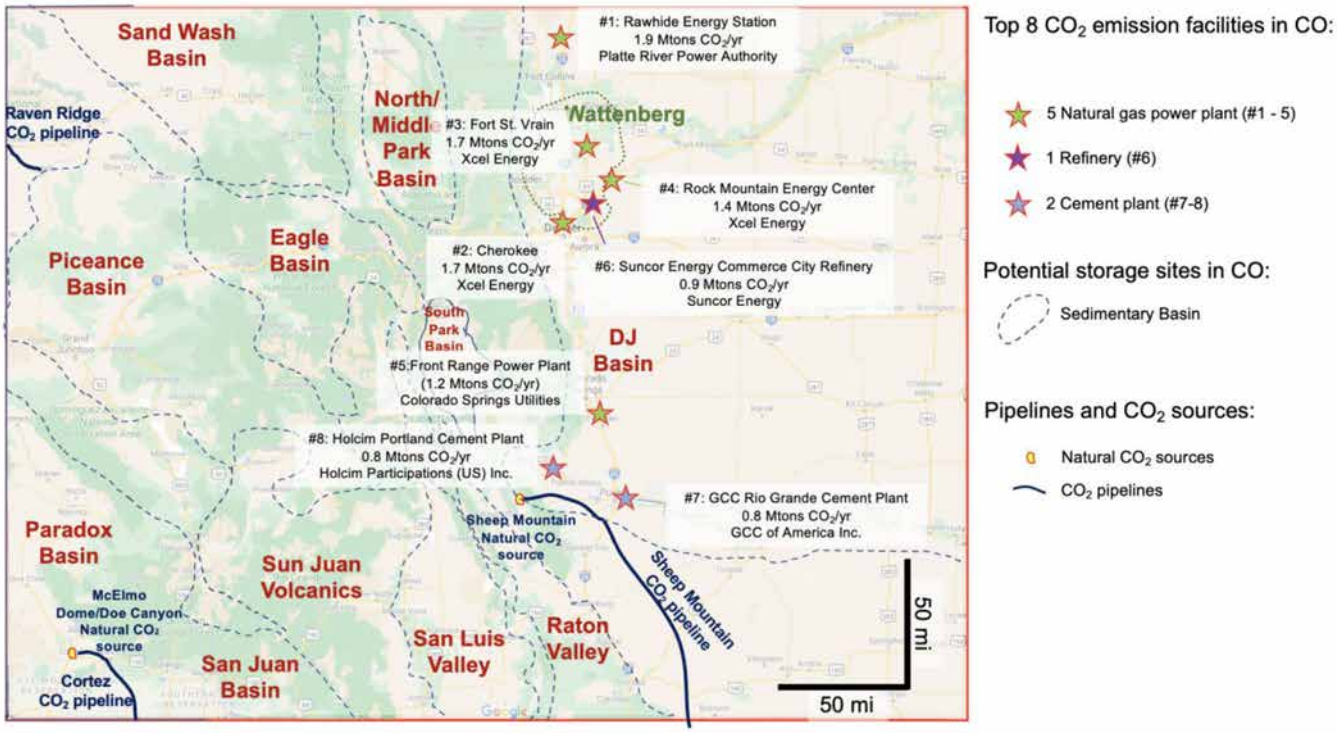

Figure 1—The 8 CO_2 top producers, CO_2 transportation pipelines, and CO_2 sinks in Colorado. All the top 8 producers are in the vicinity of DJ Basin, which makes DJ Basin a favorable storage carbon due to the vicinity.

The stacked formations of oil/gas reservoirs and saline aquifers in the DJ Basin are potential CO_2 storage formations (Young et al., 2007; Drake et al., 2014). The generalized stratigraphic column of DJ Basin is presented in Fig. 2. The color with red background represents the potential carbon storage formations, while the blue color shows the regional seals. The current drilling activities with multistage fracture stimulation of horizontal wells target the oil/gas producing zones of Niobrara through Greenhorn. All these oil/gas producers penetrate the shallower Pierre sandstone formation. This study will quantify the carbon leakage risk when the Pierre sandstone is the carbon storage formation and when one nearby Niobrara legacy well penetrates the storage formation and may behave like potential leakage pathway.

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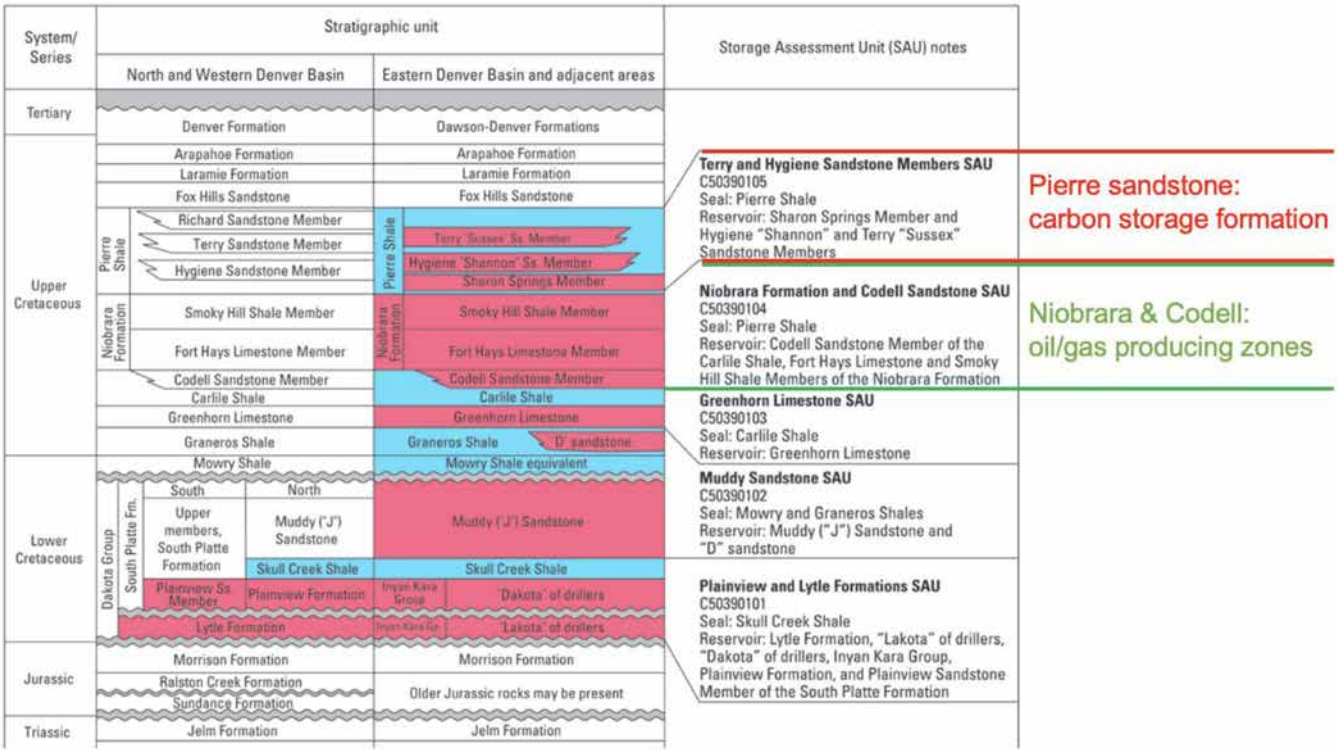

Figure 2—Generalized stratigraphic column of the DJ Basin. The red highlighted background represents reservoirs that are potential carbon storage units, whereas the blue highlighted background indicates the regional seal (modified from Drake et al., 2014).

Leakage study in the Wattenberg Field

Previous studies in the Wattenberg field have assessed the hydrocarbon migration to freshwater aquifers, which can be used as an analogy for potential CO_2 leakage. Lackey et al. (2021a) showed that the oil and gas wells in the Wattenberg Field have been observed with lots of integrity issues and linked to multiple groundwater contamination incidents. As a result, the operators have been required to routinely test the well integrity since 2009. When compared with other major oil and gas fields in Colorado, well integrity issues occur more frequently in the Wattenberg field (Lackey et al., 2021b). Based on the high incidence of well integrity issues and availability of testing data in the Wattenberg field, Lackey et al. (2021a) analyzed the available well integrity testing data and examined the factors that influence integrity issues. It was found that well integrity is primarily influenced by well location and corresponding subsurface geology. When testing the ground around the wells for indicators of gas leakage, Lackey et al. (2021b) found that 3.0% of the Colorado wells tested show potential leakage outside the well.

Fleckenstein et al. (2015) examined 17,948 wells in the Wattenberg field of the DJ Basin and found that 10 wells exhibited signs of hydrocarbon migration to freshwater aquifers. The commonality is that all the 10 wells have a shallow surface casing, which is set at the depth of 417 – 590 ft, as marked by the red box in Fig. 3. The surface casing depth here is much shallower than the bottom depth of the freshwater aquifer, which is 763 ft. The main migration path is the annulus between the outside of the casing and the wellbore. It was also discovered that the wells with different ages, cement types, and barriers, will have different levels of failure potential.

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Table 1: Vertical well barrier summary.

	BARRIERS	BARRIER TYPE	DESCRIPTION
CATEGORY 1	1	HYDROSTATIC PRESSURE	VERTICAL WELL, SHALLOW SURFACE & TOC NOT ABOVE TOP OF NIOBRARA
CATEGORY 2	1	HYDROSTATIC PRESSURE	VERTICAL WELL, SHALLOW SURFACE & TOC NOT ABOVE TOP OF SUSSEX
CATEGORY 3	2	PRODUCTION CASING + HYDROSTATIC PRESSURE	VERTICAL WELL, SHALLOW SURFACE & TOC ABOVE TOP OF SUSSEX
CATEGORY 4	2	PRODUCTION CASING + CEMENT COLUMN	VERTICAL WELL, SHALLOW SURFACE & TOC ABOVE SURFACE CASING SHOE
CATEGORY 5	3	SURFACE CASING + CEMENT + HYDROSTATIC PRESSURE	VERTICAL WELL, DEEP SURFACE & TOC NOT ABOVE TOP OF SUSSEX
CATEGORY 6	3	SURFACE CASING + CEMENT + PRODUCTION CASING	VERTICAL WELL, DEEP SURFACE & TOC ABOVE TOP OF SUSSEX
CATEGORY 7	4	2 CASING STRINGS + 2 CEMENT COLUMNS	VERTICAL WELL, DEEP SURFACE & TOC ABOVE SURFACE CASING SHOE

Table 2: Vertical well barrier failure summary.

	WELL COUNT	POSSIBLE BARRIER FAILURES	% FAILURE	CATASTROPHIC BARRIER FAILURES	CATASTROPHIC FAILURE %	AVG AGE OF WELL	P&A WELL COUNT	CURRENT WELL COUNT	AVG SURFACE DEPTH (FT)	AVG TOP OF CEMENT (FT)
CATEGORY 1	399	92	23.06%	3	0.75%	1986	125	114	417	7,296
CATEGORY 2	7,811	276	3.53%	6	0.08%	1994	738	5,446	476	6,022
CATEGORY 3	3,407	20	0.59%	1	0.03%	2007	95	4,548	590	2,719
CATEGORY 4	1,063	0	0.00%	0	0.00%	2008	8	1,289	572	417
CATEGORY 5	1,374	13	0.95%	0	0.00%	2000	112	916	967	6,473
CATEGORY 6	2,069	0	0.00%	0	0.00%	2007	24	2,654	911	2,950
CATEGORY 7	705	0	0.00%	0	0.00%	2010	1	758	941	710
TOTAL	**16,828**	**401**	**2.4%**	**10**	**0.06%**		**1103**	**15,725**		

Figure 3—A study based on wells in the Wattenberg Field of DJ Basin shows that the well barrier failure rate is related to well ages, depth of surface casing, and barriers (modified from Fleckenstein et al., 2015).

Additionally, the surface casing pressure (SCP) can be used as an indicator of well integrity loss and gas migration. Lackey et al. (2017a) conducted an SCP test on 3923 wells in the Wattenberg Field and found that 270 wells (around 7%) showed leakage potential, which is mainly due to the low depth of the surface casing. Newer horizontal wells pose a lower leakage risk because they were predominantly constructed with deeper surface casings. Based on the well analysis in the Wattenberg field, Lackey and Rajaram (2017b) also demonstrated that the uncased wells show greater integrity loss than the fully-cased wells. Accordingly, leakage potential performed in this study will be classified by multiple factors including well cement type and well barriers, which is achieved by changing the wellbore types and varying the values of effective well permeability.

Permeabilities in the disturbed zone

The "disturbed zone" is the region outside casing through the damaged rock zone. It is challenging to estimate a reasonable range for the effective well permeability of the "disturbed zone". Multiple simulation work and field experiments have proposed different approaches. Gasda et al. (2008) proposed a downhole pressure test that can be performed as a field experiment. The effective wellbore permeability was measured within a low-permeability caprock, bounded above and below by permeable reservoirs, by pressurizing the reservoir below and measuring the response in the reservoir above. Numerical simulation was leveraged for relating the pressure response to effective well permeability. This study was able to identify the effective well permeability across a wide range of values, where the effective well permeability is in the same order as, or greater than, the reservoir permeability and the well permeability dominates the leakage. However, if the well permeability is significantly lower than the reservoir permeability but still higher than the caprock permeability, the effective well permeability cannot be identified.

To quantify the effective well parameters that control vertical fluid migration along the exterior of the casing, the first field test called the "vertical interference test (VIT)" was performed on existing wellbores in 2006 (Gasda et al., 2013). The estimated wellbore permeabilities from three datasets range from 1.7 mD to 170 mD, which are at least 2 orders of magnitude greater than that of the cement alone. This indicates

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that the poor bonding at the casing-cement or cement-rock interface, instead of the cement itself, are likely the main flow leakage conduits through the barrier system.

Tao et al. (2014a) estimated the effective well permeability range using field observations of wells that exhibit sustained casing pressure (SCP). Depending on the slow/fast buildup rate of the casing pressure and a simple CO_2-leakage model, the effective well permeability was found to be in the range of micro-darcies (μD) to milli-darcies (mD). By assuming that the gas migration pathway through leaky wellbores is analogous to the CO_2 leakage path, Tao and Bryant (2014) analyzed 300 leaky wellbores in 6 fields and estimated the well permeabilities along the leaky wellbores to be between 10 μD and 10 mD.

However, there is a gap between the leakage potential of different ages for wells versus the effective well permeability. Because of the lack of such data, a large range of effective well permeabilities (0.01mD, 1mD and 100 mD) are used in this study to estimate the leakage rate.

Methodology

Models based on the reservoir simulator CMG and the tool NRAP-IAM were built to evaluate the leakage rate through an existing well. CMG is an industry standard reservoir simulation software developed by the Computer Modelling Group (CMG). NRAP-IAM is an open-source tool released by DOE's National Risk Assessment Partnership, and it is able to address the key risk performance questions at geologic carbon storage sites across a project's life cycle (NETL, 2022).

The CMG model simulates the pressure and saturation changes during and after CO_2 injection. The results are then converted to look-up tables and imported to the NRAP-Open-IAM tool to simulate CO_2 leakage from the storage formation. NRAP-Open-IAM contains multiple components that estimate the fluid leakage rate through legacy wells. When coupling with a reservoir component, the CO_2 leakage from a storage formation to overlying aquifers can be calculated (Baek et al., 2021). In this research, the multi-segmented wellbore model and the open wellbore model components are simulated and the results are compared.

The multi-segmented wellbore component allows for the segmentation of the legacy wells penetrating the overlying stratigraphy into several intervals to simulate a site's specific stratigraphic and hydrogeologic properties. Baek et al. (2021) validated the analytical model against numerical reservoir flow simulations and found that the model is able to accurately predict the transport of two-phase fluids (brine and injected CO_2) through the well over time.

Bacon et al. (2021) introduced the Open Wellbore component of NRAP-Open-IAM, which is a non-isothermal and multiphase model that integrates wellbore and reservoir flow of CO_2 and variable salinity brine, intending to calculate CO_2 and brine leakage rates into a shallow underground source of drinking water (USDW) aquifer and to the atmosphere (Pan et al., 2011). Please note that the open wellbore model assumes that the wellbore is completely open, indicating that the annular space outside the casing completely lacks cement or other material. Due to the assumption that the open borehole passes through the storage formation and connects to the shallow drinking water aquifer layers, unrealistically high CO_2 leakage rates may be generated. However, this assumption is consistent with EPA's guidance for calculating the Area of Review. (White et al., 2017)

Results

CMG model setup

A one-square-mile layer cake reservoir simulation model was created in CMG (Fig. 4). The CO_2 injector is set up at the center of the section, while the other legacy well (Well A) that might be a potential leakage pathway is about 1100' ft away. CO_2 injection was simulated into the storage formation of Pierre sandstone, which has a thickness of 2000 ft, a porosity of 14%, and permeability of 2-8 mD. The model is run for 130

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years in total: 30 years injection with an injection rate of 1 MMton/year, and 100 years post-injection. The properties of the one-square-mile study area in the DJ Basin are presented in Table 1.

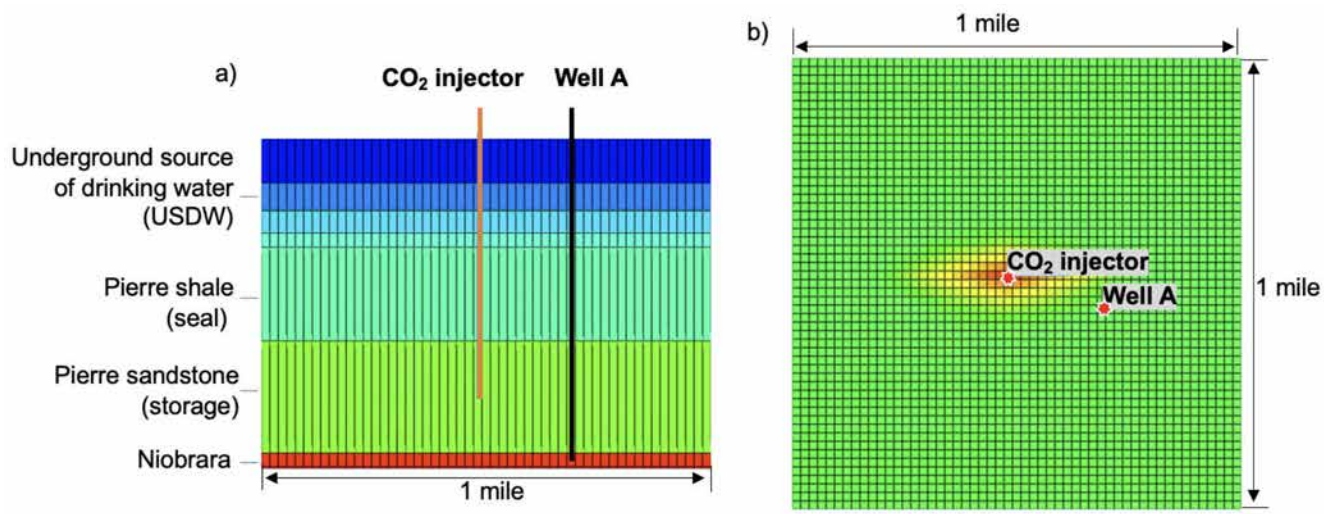

Figure 4—Reservoir simulation model in CMG. a): the side view shows that the CO_2 injector injects CO_2 into the storage formation, while the Well A was drilled in the oil-producing formation of Niobrara; b): the top view shows that the CO_2 injector is in the center of the study area, and Well A is about 1100' away.

Table 1—Formation properties in the study area

Formation Type	Formation Name	Top (ft)	Thickness (ft)	Porosity (%)	Permeability (mD)
Shale3	Shale	170	30	8	0.1
USDW/Aquifer2	Dawson	200	1000	32	200
USDW/Aquifer2	Denver	1200	600	31	200
Shale2	Shale	1800	20	8	0.1
Aquifer1	Arapahoe	1820	500	30	200
Aquifer1	Larimer-Fox Hill	2320	320	32	200
Shale1	Pierre Shale	2640	2105	10	0.1
Storage formation	Pierre Sandstone	4745	2522	14	2 - 8
Reservoir formation	Niobrara	7267	300	8	0.0001
Reservoir formation	Codell	7567	50	12	0.001

NRAP-IAM model simulation

The tool of NRAP-IAM is used to estimate the carbon leakage through Well A and to evaluate the effect of leakage on the USDW layer. The reservoir simulation results, together with the geologic background and well settings, are the key inputs of the NRAP tools. The pressure and saturation change for each grid cell are exported from the reservoir simulation model, and then imported into the NRAP-IAM tool. The geological information includes stratigraphy and properties of the formations above the carbon storage formation. Three shale intervals and two aquifers were simulated above storage target formation Pierre Sandstone, as listed in Table 1.

Sensitivity analysis has been done to investigate the effect of different parameters on the leakage rate. Here, we tested 3 scenarios. Scenario 1 is the case when the permeability of the carbon storage formation changes and the leakage wellbore is multi-segmented; Scenario 2 is when the permeability of the carbon storage formation changes but the leakage wellbore is open; Scenario 3 is when the storage formation permeability is fixed but the effective well permeability along the seal formation changes.

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Scenario 1: Multi-segmented wellbore case study. Scenario 1 simulates the case when the leakage wellbore is multi-segmented. Due to the heterogeneity of the Wattenberg field, different values are set for storage formation permeability: 2 mD, 4 mD, 6 mD, and 8mD. In Fig. 5, we present the CO_2 leakage rate and CO_2 leakage ratio over time when the storage formation permeability changes. The leakage rate refers to the tons of CO_2 leaks at each time step, which has a unit of tons/year; the leakage ratio is defined by the total carbon leaked divided by the total injected carbon at each time step.

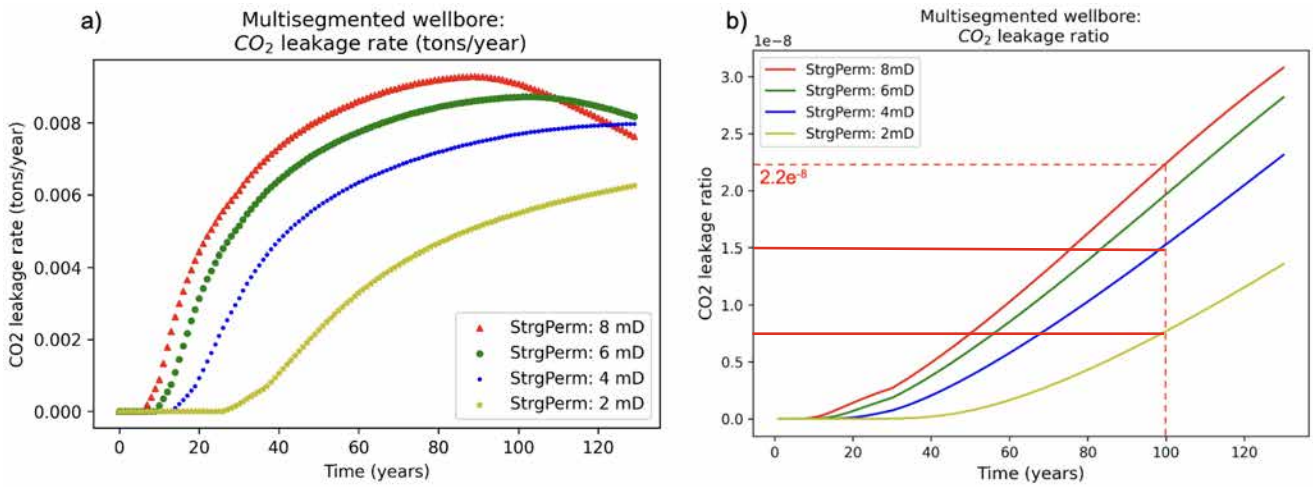

Figure 5—Multi-segmented case study: a) CO_2 leakage rate increases as the permeability of storage formation increases; once a certain point is reached, the leakage rate decreases; b) The CO_2 leakage ratio, which is referred to as the leaked CO_2 amount divided by the total injected CO2 amount, increases over time.

The CO_2 leakage rate with time is shown in Fig. 5 a). When the formation permeability decreases from 8 mD (red dotted curve) to 2 mD (yellow dotted curve), CO_2 leakage occurs later, which is about 7 years, 10 years, 15 years, and 25 years after the injection starts. When focusing on the red dotted curve (formation storage permeability = 8 mD), we can observe that the leakage rate reaches the highest around year 90, then decreases over time. For the green curve (permeability equals 6 mD), the highest leakage rate occurs at year 110. But for the blue and yellow curves, the leakage rate keeps increasing with time and the maximum values are not reached within the 100-year post-injection period.

As presented in Fig. 5 b), CO_2 leakage ratio increases over time, and the larger storage permeability scenario shows a higher leakage ratio. For instance, the total CO_2 injection is 30 MM tons at year 100; when the storage formation permeability is 8 mD, the total carbon leakage amount is 0.67 tons at year 100. Hence, the leakage ratio at year 100 is calculated to be $2.2*10^{-8}$, as marked in Fig. 5 b). When the simulation ends at year 130, the CO_2 leakage ratio is approximately $3*10^{-8}$, indicating that only a small portion of the injected CO_2 leaks via the multi-segmented wellbore after 100 years of post-injection.

Scenario 2: Open borehole case study. The leakage results of open wellbore type are investigated here. The open wellbore scenario represents the wells in the Category 1&2 in Fig. 3. These wells were drilled before 1994 and have a surface casing depth of 476 ft or shallower. The only barrier is the hydrostatic pressure. The leakage rate over time is shown in Fig. 6 a). When compared with the multi-segmented wellbore scenario in Fig. 5 a), it can be observed that the leakage starting time varies little by the wellbore type, but the leakage rate values are significantly impacted by the type of wellbore. The leakage rate for the open wellbore scenario (Fig. 6) is approximately 10^6 times greater than the multi-segmented wellbore scenario (Fig. 5). For the four cases with varying storage formation permeability values (8 mD, 6 mD, 4 mD, 2 mD), the maximum leakage rate is observed at years 35, 40, 55, and 100, respectively. This is much earlier than that observed for the multi-segmented borehole case study (in Fig. 5). Another notable difference is that after

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the maximum leakage rate is reached, CO_2 leaks at a relatively constant rate, indicating the necessity of continuously monitoring CO_2 leakage.

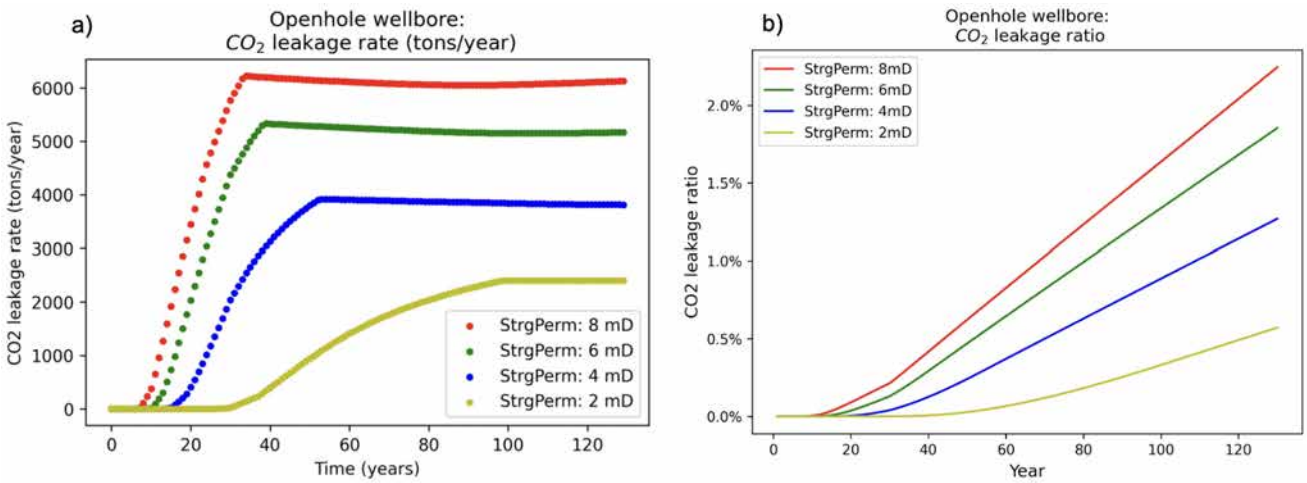

Figure 6—Open borehole case study: a) CO_2 leakage rate increases over time, and then keeps constant after a certain time point; b) the CO_2 leakage ratio increases through time. When the storage permeability is 8 mD, the leakage ratio reaches 2.5% at year 130.

The CO_2 leakage ratio over time is shown in Fig. 6 b). Because CO_2 leaks at an initially increasing but then flat rate, the leakage ratio keeps increasing. At year 130, the leakage ratio is as high as 2.5%, meaning that 2.5% of the total injected CO2 has leaked. The continuing growth trend of leakage ratio in Fig. 6 b) implies that more CO_2 will leak via the open wellbore as time increases.

Scenario 3: Change of effective well permeability for the multi-segmented wellbore. When the storage formation permeability is 4 mD and the leakage wellbore is set as multi-segmented, but all other parameters (Table 1) are kept unchanged, we analyzed the impact of effective well permeability on leakage results. As mentioned earlier, the well permeability is related to the age and completion type of the legacy well. This study uses a wide range of effective well permeability: 0.01 mD, 1 mD, and 100 mD. As presented in Fig. 7 a), b) and c), the same shape of the CO_2 leakage rate curve is observed when the effective well permeability varies. The only difference among Fig. 7 a) - c) is the values of the leakage rate in the vertical axis, demonstrating that the carbon leakage amount is linearly related to the effective well permeability. This suggests the significance of determining the most reasonable value or the magnitude level of the effective well permeability.

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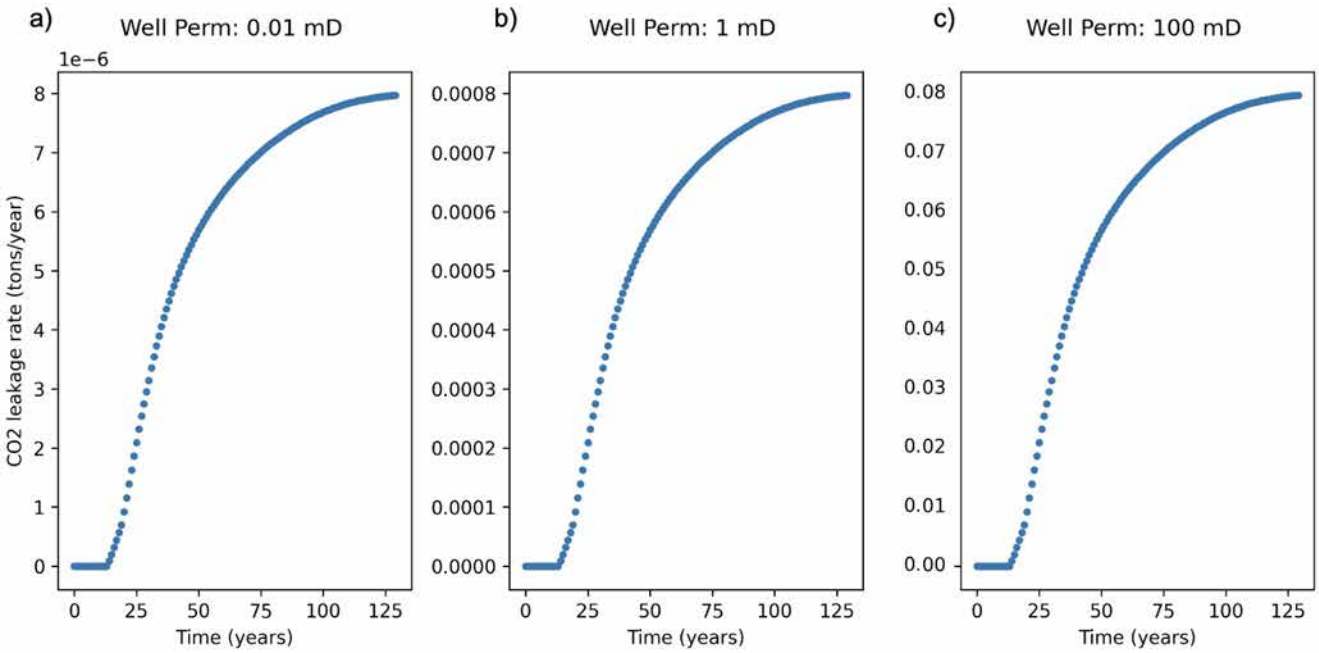

Figure 7—Case study based on varying values of effective well permeability. The carbon leakage rate is linearly related to the value of effective well permeability.

Conclusions and Discussion

This research delves into a topic that is critical but poorly studied for CCUS applications. It can be applied to guide the selection of potential carbon sequestration geological sites and the appropriate placement of CO_2 injection wells. It also provides insights into legacy well remediation for the protection of drinking water aquifers. Specifically, the following conclusions are drawn from this research.

- Carbon leakage varies significantly by the type of wellbore. For the multi-segmented wellbore, only a small portion of the injected CO_2 leaks via the legacy well that is 1100' away from the carbon injector. However, when the wellbore type is set as open, the leakage ratio reaches 2.5% at the end of the 130-year simulation period.
- I The permeability of the storage formation affects the carbon leakage rate. The formations with higher permeability values correspond to earlier leakage time and higher leakage rate. The maximum leakage rate may not be reached until 100 years after CO_2 injection, indicating that continuous monitor may be needed to last for 100 years or longer.
- The open wellbore behaves as a leakage conduit. When the storage formation is 8 mD, the open wellbore leads to the leakage rate of 6000 tons/year from year 35 to 130. Considering the leakage rate is relatively constant and no decreasing trend is observed over time, a much higher leakage ratio will be expected as the simulation time lasts longer. It suggests that when selecting the geological sites for carbon storage, the open wellbore should be avoided or the expense to treat it should be considered in economic calculations.
- The Wattenberg Field in the DJ Basin is a good candidate for carbon storage because it is near major CO_2 producers, and it has stacked formations for carbon storage and a good seal. However, pre-existing oil/gas wells in this area can be potential leakage pathways for CO_2 leakage during and after the geological carbon storage operations.
- When relating our observations to a previous leakage study in the Wattenberg field (Fleckenstein et al., 2015), a good storage site in this field should not include the wells in Category 1&2, which are the wells drilled before 1994 and have a surface casing depth of 476 ft or shallower. If these wells

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are in the potential storage site and cannot be excluded, we may need to plug or treat these wells before CO_2 injection starts.

– The leakage rate varies significantly by effective well permeability, which is determined by well age and completion type. Specific field tests will be necessary to investigate this parameter and to reduce the uncertainty in the leakage estimates.

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